

MUHAMMAD USMAN SHAMS

Software Engineer (Senior UI/UX Designer x AI Research Engineer)

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PROFESSIONAL SUMMARY

Software Engineer, Specialized as AI Researcher, and UI/UX Designer with a demonstrated record of delivering production-grade AI systems and internationally published research. Completed BS Software Engineering (CGPA 3.68/4.00, Rank 4th / Batch 2022–2026) at KFUEIT while concurrently serving as AI Researcher at Beijing Institute of Technology and Senior UI/UX Designer at GTech Solutions, Australia. Author of 4 peer-reviewed journals spanning deep learning, digital twin systems, and healthcare AI. Invited presenter at IEEE EIC International Conference 2025 and recipient of the Dean's Honor Certificate for Outstanding Semesters Academic Achievement. Expertise spans AI/ML pipeline engineering, digital twin architecture, Systems Designs for Complex Products, and full-cycle product design.

EDUCATION

BS Software Engineering

2022 – June 2026

Khwaja Fareed University of Engineering & Information Technology (KFUEIT), Pakistan

CGPA: 3.68 / 4.00 · Class Rank: 4th / Batch 2022–2026 · Dean's Honor Certificate — Outstanding Semesters Academic Achievement

Intermediate — Bio Sciences

2019 – 2021

Forman Christian College, Lahore, Pakistan

Grade: 86% - Grade A

PROFESSIONAL EXPERIENCE

Senior UI/UX Designer

2025 – 2026

GTech Solutions · Sydney, Australia · Remote

- Lead end-to-end design of next-generation HRM platform — branding, system design, high-fidelity web and mobile UI
- Define product vision, strategy, and roadmap in collaboration with developers, HR specialists, and product managers
- Produce wireframes, prototypes, and functional designs; run user testing cycles and iterate based on quantitative feedback
- Create pitch decks, executive reports, corporate websites, and full brand identity including logos and design guidelines

Software Engineer — AI Researcher

2024 – Present

Beijing Institute of Technology · Beijing, China · Project Base

- Completed 9+ machine learning research projects in under 16 months under Dr. Muhammad Usman Shoukat's supervision
- Authored 4 peer-reviewed publications spanning bearing fault diagnosis, adversarial ML, digital twin healthcare, and energy harvesting; invited presenter at IEEE EIC International Conference 2025
- Developed novel deep learning architectures for SOH/RUL battery prediction, KAN-based autonomous vehicle digital twin communication, and multi-modal anomaly detection
- Collaborated with international research teams across Pakistan and China, managing data pipelines, model training, and academic writing concurrently with studies

Design Lead — UI/UX Designer

2022 – 2023

Shingle · Lehi, Utah, USA · Remote

- Owned the complete design lifecycle for law firm client websites — discovery, wireframes, prototypes, and developer handoff
- Conducted user research, built personas, mapped user journeys, ran usability tests, and iterated on feedback
- Translated client strategy into elegant, human-centric digital experiences on time and within budget

SKILLS

Languages & Frameworks: Python · HTML / CSS · JavaScript ·

AI / ML: Keras · TensorFlow · PyTorch · BERT / HuggingFace · LSTM · GRU · KAN · SHAP · RAG · LLMs

Backend & Infra: FastAPI · Flask · WebSocket · REST API · VPS · Server

Design Tools: Figma · Adobe XD / AI / PS · Sketch · Canva · MS Designer

Design Disciplines: UI/UX Design · Product Design · Branding · Design Systems · UX Research

Research Areas: Digital Twin Technology · Agentic AI · Intelligent Systems · Machine Learning AI & IoT · Industry 4.0 / 5.0

Dev Tools: VS Code · Git / GitHub · Google Colab · Jira · FileZilla

Soft Skills: Cross-Cultural Collaboration · Problem Solving · Technical Writing · Remote Team Leadership · Time Management

Languages: Urdu / Punjabi (Native) · English C1 (Proficient)

AI PROJECTS

FYP: Behavioral Agentic AI — Sentiment-Aware Escalation System

Agentic AI Product Solving Global E-commerce Customer Care Service Problem - VPS Deployed

A full-scale AI customer support platform that understands customer emotions, responds intelligently in 12+ languages, and automatically escalates frustrated customers to human agents. The system reads sentiment, retrieves relevant knowledge, generates responses, scores escalation urgency, and updates a live dashboard — all in one seamless pipeline. Deployed on a live server with secure login, churn prediction, and satisfaction tracking built in.

↳ · 19 supported languages · Live on VPS

FBG Battery Monitoring System

Using Fiber Bragg Grating (FBG) sensors for Lithium-ion Batteries - Research

A real-time health monitoring system for lithium-ion batteries using fiber-optic sensors embedded in the battery casing. An AI model trained on 731,000+ data points predicts battery charge level with 93% accuracy and detects 6 types of failures — including thermal runaway and swelling — before they become dangerous. Results are visualized on a live web dashboard with explainable AI showing exactly which sensor readings drove each prediction.

↳ 92-93% Accuracy · SHAP Explainability · Live Dashboard

Multi-task KAN — Autonomous Vehicle Digital Twin

Multi-Task Autonomous Vehicle System · Kolmogorov-Arnold Networks · Digital Twin · Multi-modal - Research

A new type of AI model designed to detect multiple faults simultaneously in self-driving car systems — using a virtual replica (digital twin) of the vehicle for safe testing. The model processes data from cameras, radar, and LiDAR together, outperforming standard neural networks on combined fault detection tasks.

↳ Comprehensive Research Project

Adversarial Attack Defense Framework

Defense Framework for Adversarial Attack

A defense system that protects AI image classifiers from deliberate sabotage. Attackers can subtly alter images to fool AI models — this project builds a two-layer defense that strips out the hidden interference and restores correct classification. Achieved 92% accuracy on corrupted images. Published in SJER 2025.

Smart Pharma — Intelligent Inventory System

Featured Intelligent Inventory System Using Advanced Sensors & Analytics - Research

An end-to-end AI system for managing pharmaceutical supply chains — from factory floor to final delivery. The platform tracks inventory, monitors product quality, predicts shortages before they happen, and flags delays early. Designed the complete user interface from wireframes to final handoff.

Cancer Tumor Classification — Multi-Model Benchmark

Application of Multiple Modals on a Single Data-Points for Comprehensive Modals Evaluation.

A systematic comparison of 9 deep learning models for classifying tumors as benign or malignant — spanning CNNs, transformers, and hybrid architectures. A custom-built HybridNet model achieved the highest accuracy (94.57%) while also training the fastest, demonstrating that thoughtful architecture design can outperform larger, more complex models for medical diagnosis.

↳ 9 models evaluated · 94.57% accuracy · ROC AUC 0.9502

Semi-supervised Learning — Noise & Imbalance Research Framework

Solved Critical Problems in SSL for Better Modal Performance on Highly Clustered data Points - Research

A research framework solving two core problems in training AI on limited labeled data: (1) incorrect auto-generated labels that pollute the training process, and (2) heavily skewed class distributions that cause models to ignore rare categories. The proposed HybridBalance approach combines graph-based label filtering with prototype-guided class balancing, improving class accuracy by 45% and 15% for the Complete Data points including Majority & the Minority Classes.

↳ Unified Solution Framework

PEER-REVIEWED PUBLICATIONS

PUBLISHED JOURNALS

- P1 Sarwar A., Sarwar D., Shams M.U., Islam M., Afzal M.J., & Yasir A.B. "Rolling Bearing Fault Detection using Deep Learning Model: Intelligent Manufacturing Industry 4.0/5.0." Scientia: Technology, Science and Society, 1(3), 104–113, 2024. DOI: 10.59324/stss.2024.1(3).06
- P2 Khan M.R.A., Fareed B., Khan M.F.A., Nisar M., Shams M.U., & Rahmani A.N. "Design and Implementation of a Smart Footsteps Power Generation System." European Journal of Applied Science, Engineering and Technology, 3(5), 171–181, 2025. DOI: 10.59324/ejaset.2025.3(5).14
- P3 Nisar M., Fayyaz N., Ahmed M.A., Shams M.U., & Fareed B. "Incremental Development of a Framework for Mitigating Adversarial Attacks on CNN Models." Scientific Journal of Engineering Research (SJER), 1(4), 2025. DOI: 10.64539/sjer.v1i4.2025.349
- P4 Din G.M., Shams M.U., Rahmani A.N., Rehman A., Ahmed M.A., & Shoukat M.A. "Comparative Analysis on AI-Driven Human Digital Twin for Personalized and Predictive Medicine." Journal of Clinical Practice and Medical Research, 1(1), 30–38, 2025. DOI: 10.59324/jcpmr.2025.1(1).06

AWARDS, HONOURS & CERTIFICATIONS

AWARDS & HONOURS

Dean's Honor Certificate for Outstanding Semester Academic Achievement August 2025

Faculty of Electrical and Computer Engineering, KFUEIT, Rahim Yar Khan

Certified by Prof. Dr. Muhammad Sagir, Dean of All Faculties (No. KFUEIT/DEAN/2025/DHR/1606). Awarded in recognition of outstanding Semesters academic achievement — BS Software Engineering (Reg. SWEN221101044).

Rank 4th — BS Software Engineering Batch 2022–2026 2022 – 2026

Khwaja Fareed University of Engineering & Information Technology (KFUEIT)

Achieved 4th rank in the entire BS Software Engineering batch of 2022–2026, maintaining a CGPA of 3.68/4.00 across all semesters while simultaneously conducting international research.

HEC PMYP Laptop Award — Academic Excellence March 2023

Higher Education Commission of Pakistan (HEC)

Awarded by HEC Pakistan to high-performing students at premier engineering universities. Recognized for outstanding academic performance in support of exceptional student achievement.

CERTIFICATIONS

Google UX Design Professional Certificate Feb 2023

Google · User Experience (UX), UX Research, Design Systems

Principles for UX Design (2021) Aug 2023

LinkedIn · User Experience (UX), UX Research, Product Thinking

Designing for Developers Aug 2023

LinkedIn · Proactive Design Thinking, Developer-Designer Collaboration

Design Thinking: Understanding the Process Aug 2023

LinkedIn · Design Thinking, Product Development

Design Thinking — Testing and Refining Aug 2023

LinkedIn · Design Thinking, Iterative UX Process

VOLUNTEERING

Volunteer Design Writer

2023 – Present

[Muzli — Design Inspiration \(Medium\)](#) · Writer

- Share design knowledge freely with the global community — articles on UI/UX principles, tools, and creative processes
- Profile: mushams.medium.com

REFERENCES

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